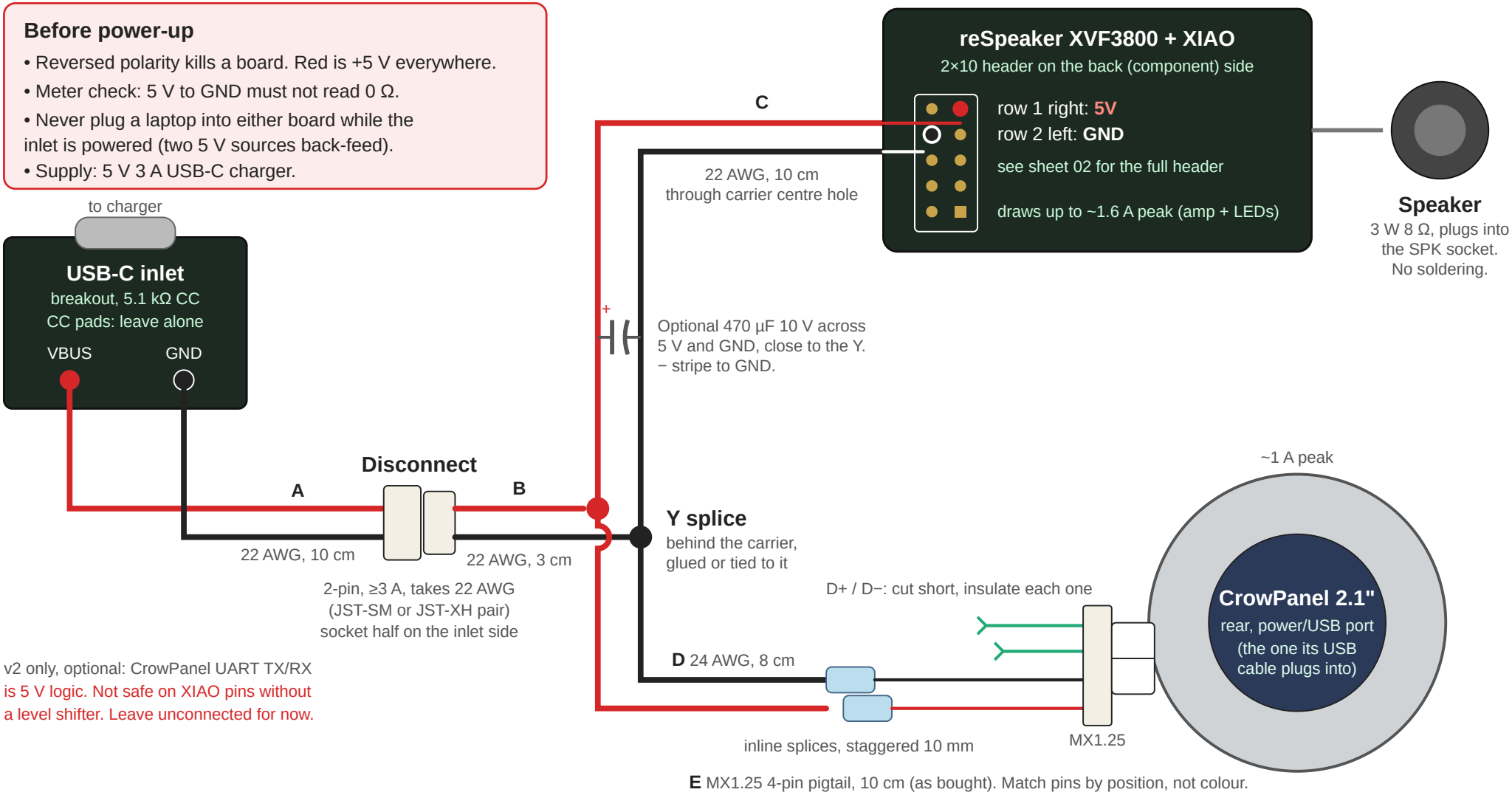
Voice Clock: power wiring

One 5 V USB-C inlet feeds both boards. Red = +5 V, black = GND. Lengths are cut lengths; cut long and trim at the test fit. Not to scale.

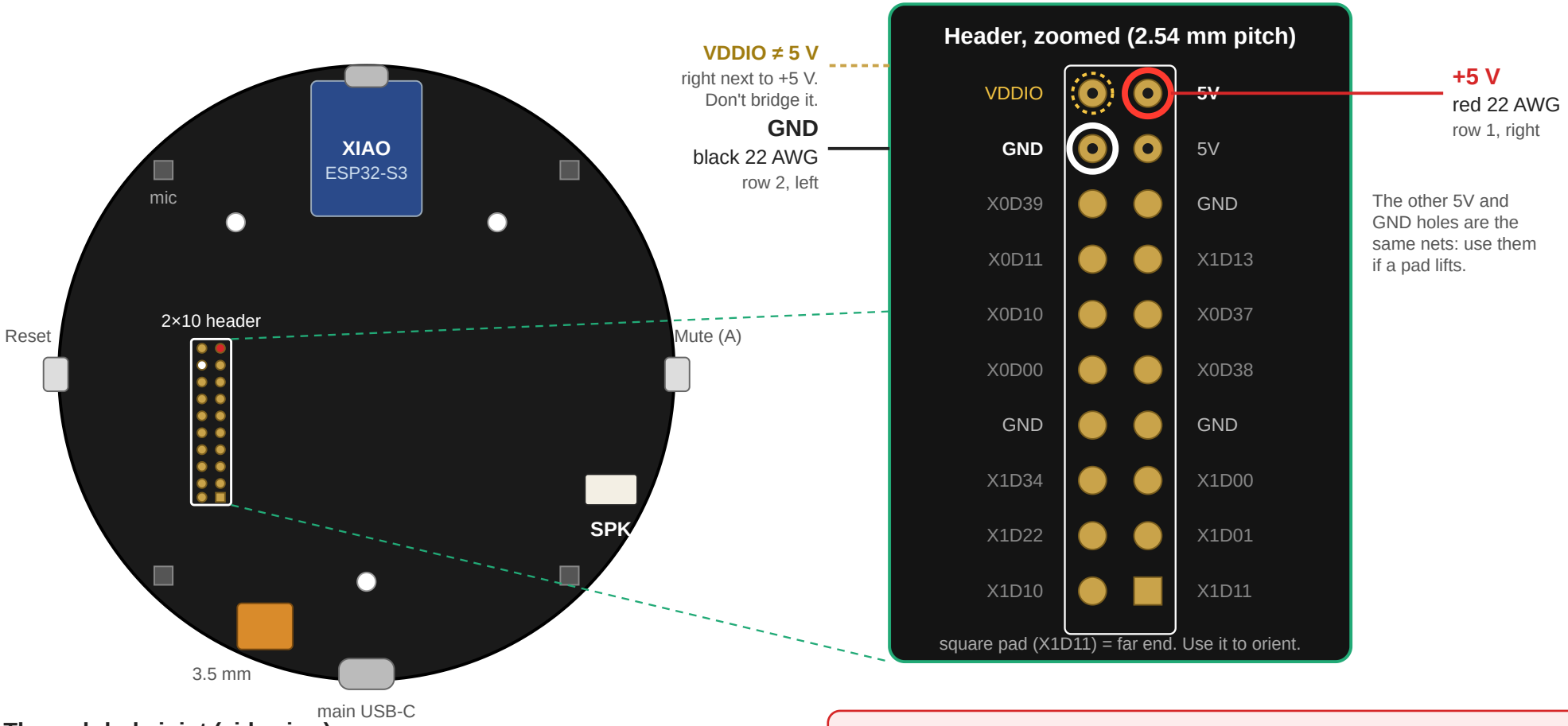
Before power-up

- Reversed polarity kills a board. Red is +5 V everywhere.
- Meter check: 5 V to GND must not read 0 Ω.
- Never plug a laptop into either board while the inlet is powered (two 5 V sources back-feed).
- Supply: 5 V 3 A USB-C charger.

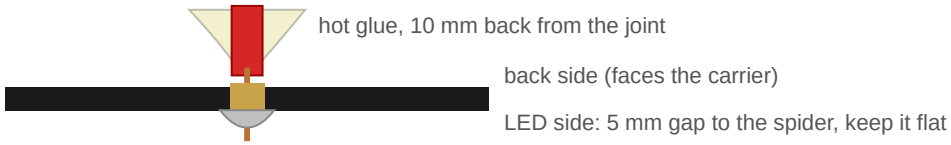


02 · reSpeaker XVF3800: 5 V and GND solder points

Back (component) side, XIAO at 12 o'clock, as in Seeed's pinout photo. Positions are approximate. Board Ø100 mm. Pin labels are from Seeed's pinout.



Through-hole joint (side view)



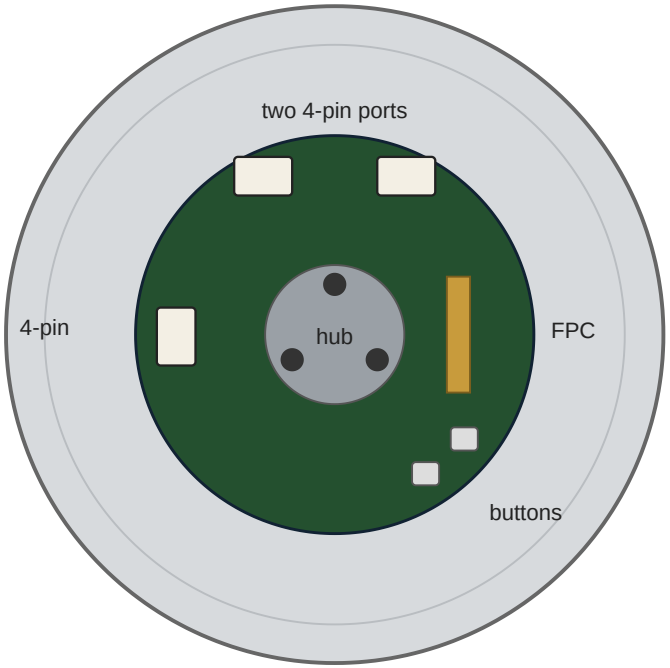
Strip 2–3 mm, tin, push in from the back, solder on the LED side, trim to ≤ 1 mm.

Do this first (2 min, bench)

1. Power the reSpeaker from its own USB-C. Meter header 5V to GND: ~5.0 V.
2. Unplug. Beep header 5V to the XIAO 5V pin: should beep (same rail).
3. GND holes sit on a ground plane: 370 °C, a chisel tip, hold 3–4 s.
4. If pins are already fitted, solder to the pin stub and heat-shrink it.

03 · CrowPanel 2.1": power through a plug-in pigtail

No soldering on the CrowPanel itself. Feed 5 V into the same 4-pin MX1.25 port that its supplied USB cable uses, with a spare pigtail, so the panel stays removable.



Rear view (layout approximate)

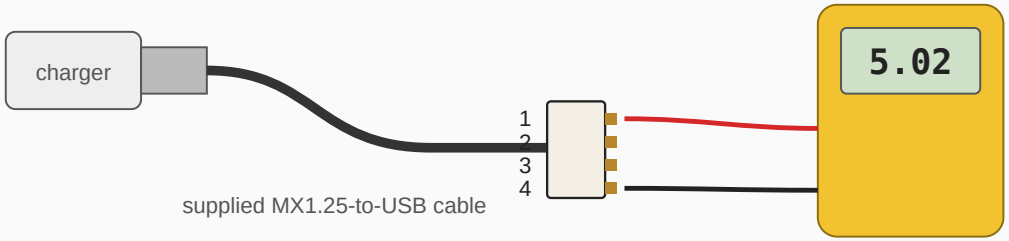
Use the port its MX1.25-to-USB cable plugs into: the one you flashed it through. Mark it with a dot.

Why not solder to the board?

Elecrow's schematic routes 5 V into the main board only through its 12-pin FPC (pins 1–2 = 5 V, 11–12 = GND) and a tiny test pad. A plug-in lead avoids heat on the display and keeps the panel removable.

Its UART pins are level-shifted to 5 V logic (v2 note).

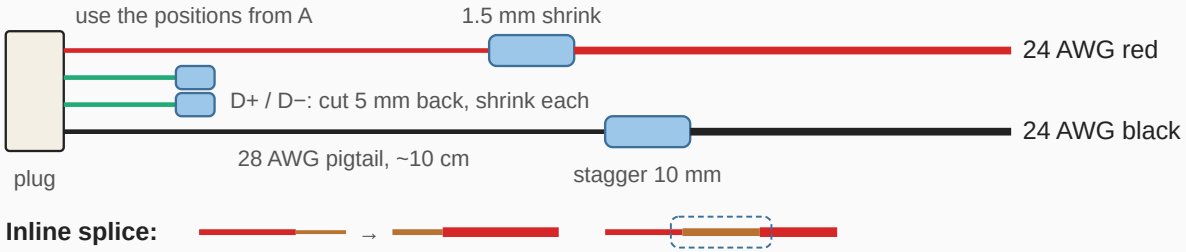
A · Find 5 V and GND on the supplied cable (meter on DC V)



Probe pairs of contacts with a needle on each probe. **+5.0 V**: red probe = 5 V, black = GND.

Write down the two positions before you go on. Pigtail wire colours are not a standard; positions are.

B · Spare MX1.25 4-pin pigtail → 24 AWG (lead D)



Slide on 1.5 mm shrink first. Strip 6 mm on each, lay them side by side, twist, solder, let cool, shrink. Then one 4.8 mm sleeve over the pair. The splices must sit behind the carrier, not in the notch.

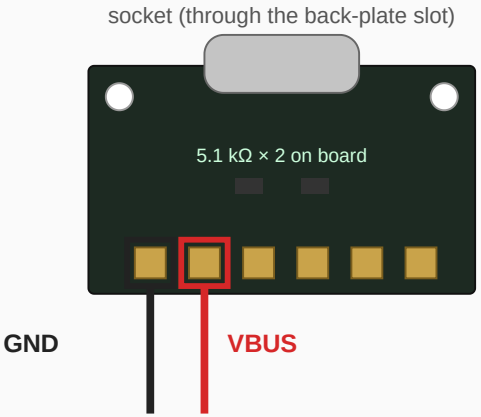
C · Test before the case

Plug the pigtail into the marked port and feed 5 V from the harness (or a bench supply set to 5.0 V, 1 A limit). The clock face should boot. If nothing happens, unplug at once and re-check step A.

04 · USB-C inlet, disconnect and Y splice

Lead A stays with the back plate. Everything after the disconnect stays with the board stack, so the back plate can come off.

USB-C breakout (22 × 13 mm)

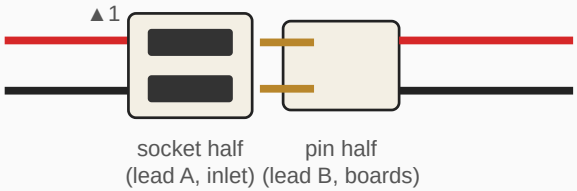


Pad order and names vary (VBUS may read V, VCC or 5V; GND may read G). Go by your board's silkscreen.

Leave CC1 / CC2 and D+ / D− alone.

Check: CC1→GND and CC2→GND each ≈ 5.1 kΩ.

Disconnect (2-pin, wire-to-wire)



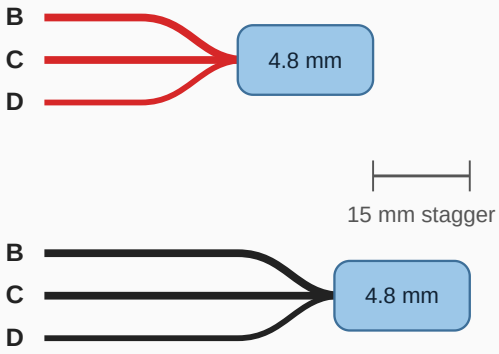
The live (inlet) side gets the shrouded sockets, so a loose lead can't short. Pin 1 = red.

Which connector

- ✓ JST-SM 2-pin (LED-strip pigtails): 3 A, 22 AWG
- ✓ JST-XH 2-pin male/female pair: 3 A, 22–28 AWG
- ✗ JST-PH: 2 A, crimps take 24–32 AWG only

Buy pre-wired pairs, then solder 22 AWG onto their leads.

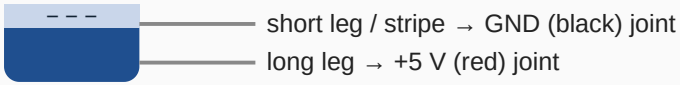
Y splice (one per colour)



B from the disconnect (22 AWG), C to the reSpeaker (22 AWG), D to the CrowPanel (24 AWG).

Slide the 4.8 mm sleeve on first. All three ends point the same way: twist, solder, trim to 8 mm, shrink.

Optional 470 µF / 10 V capacitor



Solder its legs into the two Y joints before shrinking. Keep the legs short and sleeved. It smooths the amp's current peaks.

Iron and materials

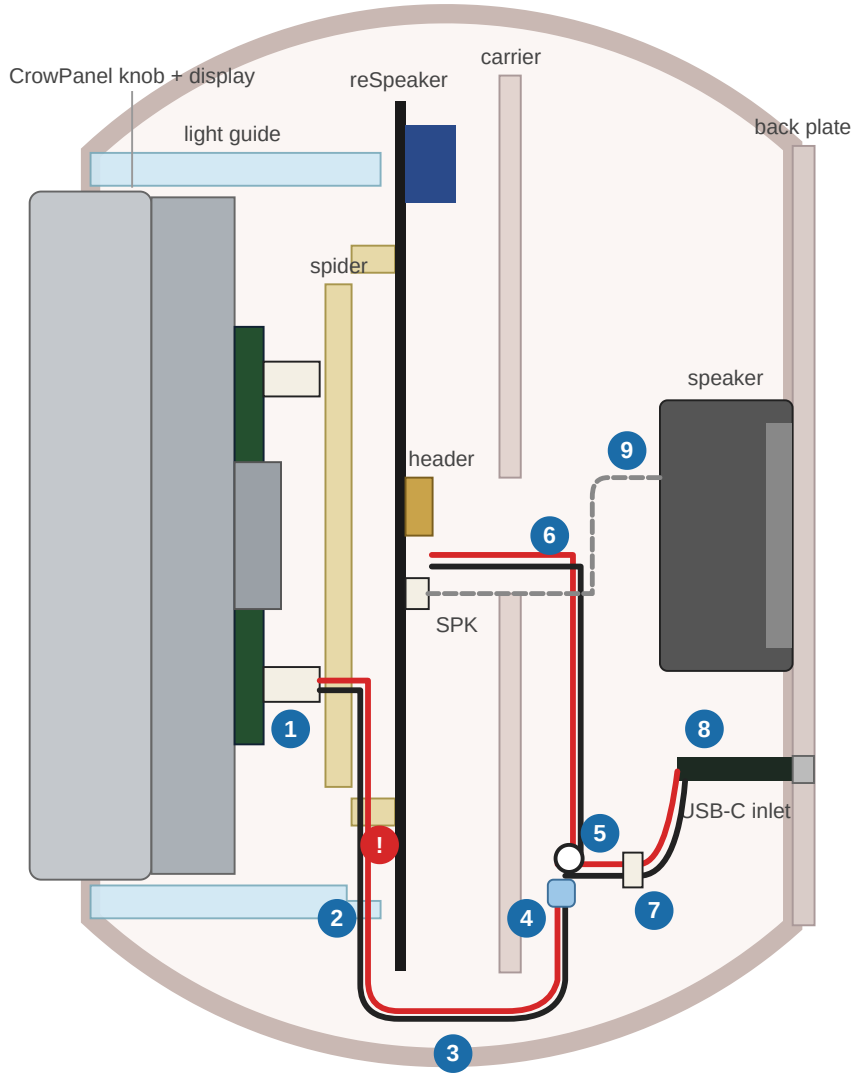
- Swap the heat-set tip for a 2 mm chisel. 340 °C for wire, 370 °C for ground-plane pads.
- 0.6–0.8 mm flux-core solder (63/37 is easiest), extra flux pen for the header.
- Silicone wire: 22 AWG red + black, 24 AWG red + black. Heat shrink 1.5, 2.5, 4.8 mm.
- Tin each stripped end before joining. Let joints cool before you move them.

Order of work

1. Bench checks: sheet 02 (reSpeaker) and sheet 03 step A.
2. Lead A: solder to the breakout, then to the disconnect socket half.
3. Lead B: the pin half's own leads, trimmed to ~3 cm.
4. Lead E + D: pigtail splice (sheet 03 step B).
5. Lead C: solder to the reSpeaker header (sheet 02).
6. Y splice: B + C + D (+ cap). Slide shrink on first!
7. Meter: 5 V to GND not 0 Ω at the disconnect pins.
8. Charger into the inlet, meter at the socket half: +5.0 V, red probe on pin 1. Then plug together: both boards boot.
9. Play audio loud: header 5V–GND stays above 4.75 V.
10. Hot-glue or tie the Y and splices to the carrier back.

05 · Where the wires run (side section)

Front (knob) on the left, back plate on the right. Parts to scale (4.5 px/mm) from the CAD, face tilt left out. Wires drawn schematically.



1. Pigtail plug in the CrowPanel power port.
2. Out through the spider, across the 5 mm gap and out through the light guide's notch (lower left from the front).
3. Over the reSpeaker's edge (3.6 mm to the shell wall) and through the carrier's wire channel.
4. Pigtail-to-24 AWG splices (E→D). Only behind the carrier, where there is room.
5. Y splice (+ optional capacitor). Hot-glue or zip-tie it to the carrier back so no tug reaches a pad.
6. Lead C through the carrier centre hole to the reSpeaker header (5V / GND).
7. Disconnect. Unplug it to lift the back plate off. Clear of the speaker and the inlet cage.
8. Lead A to the USB-C inlet. Stays with the back plate. 10 cm gives room to set the plate beside the clock.
9. Speaker lead (plug-in, no soldering) through the centre hole to the SPK socket. Zip-tie it to its anchor.

! **No joints, heat shrink or loops in the 5 mm gap** between the spider and the reSpeaker, or in the notch. Only the bare pigtail passes there (notch 4.5 × 4 mm).

Leave about 2 cm of slack on lead D so the board stack can slide into the front shell before the carrier screws go in.